Dichen Li

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Education

University of California, San Diego **09/2024-03/2026**
Master's Program in ECE department: Major in EC80 intelligent system & robotics | GPA: 3.96/4.00

University of California, Berkeley **09/2023-06/2024**
Exchange Program: Studying in EECS department
Relevant Curriculum: Designing, Visualizing and Understanding Deep Neural Networks; Introduction to Artificial Intelligence

Xi'An Jiaotong University **09/2020-06/2024**
Bachelor of Engineering: Major in Automation (Graduate with honor) | GPA: 3.92/4.30 rank: 19/220

Publications

"Towards Embodiment Scaling Laws in Robot Locomotion." **CoRL2025**
Bo Ai*, Liu Dai*, Nico Bohlinger*, Dichen Li*, Tongzhou Mu, Zhanxin Wu, K. Fay, Henrik I. Christensen, Jan Peters, Hao Su.
"Versatile Locomotion Skills for Hexapod Robots." **IROS2024**
Tomson Qu, Dichen Li, Avideh Zakhor, Wenhao Yu, Tingnan Zhang.

Awards

1st Prize of Shanxi Province, China in the 2022 National College Student Mathematical Modeling Competition

Research Experiences

Embodiment Adaptation in Robot Locomotion **04/2025-Present**
Graduate Research Intern | UC San Diego
- Trained an online locomotion controller that learns diverse gaits across multiple robot embodiments with varying continuous parameters.
- Designed and trained a transformer-based adaptation module to infer unknown robot parameters from historical trajectories, enabling robustness to motor failures, payload variations and other uncertainties, improving generalization across diverse embodiments.

Large-scale Embodiment Scaling Law **10/2024-05/2025**
Graduate Research Intern | UC San Diego
- Conducted reinforcement learning of locomotion policies on 1,000 distinct robots (hexapod, humanoid and quadruped; with different topology, geometry and kinematic features) using Isaac Lab.
- Conducted large-scale supervised teacher-student policy distillation on cross-embodiment data, and achieved real-world deployment for both teacher and student policies on Unitree Go2 and H1 robot.
- Discovered that training on accumulated embodiment improves performance, and that embodiment scaling outperforms data scaling.
- **Paper accepted to CoRL 2025 as the co-first author.** The corresponding author is Professor Hao Su.
- **Paper title: Towards Embodiment Scaling Laws in Robot Locomotion.**

Hexopod Robot Locomotion skills with Reinforcement Learning **09/2023-05/2024**
Undergraduate Research Intern| UC Berkeley
- Designed reward and trained reinforcement learning policies in complex environments for a hexapod robot with depth-sensing cameras to accomplish tasks of obstacle avoidance, stair climbing, and narrow passage traversal.
- Conducted policy distillation with the only viable observations, and deployed on real hexapod robots, achieving stair climbing, cave-like passages traversal (height reduction from 22cm to 10cm), and obstacle avoidance (maneuvering around 40x40x40cm blocks).
- **Paper accepted to IROS 2024 as the 2nd author**. The corresponding author is Professor Avideh Zakhor.
- **Paper title: Versatile Locomotion Skills for Hexapod Robots.**

Four-Wing UAV Control and Navigation **04/2022-08/2023**
Undergraduate Research Assistant | Xi'an Jiaotong University
- Developed real-time flight control firmware in C/Keil (TM4C) with multi-level PID and mission logic.
- Adapted ROS system files for sensor data management and utilized hector_mapping for UAV indoor localization. Developed a real-time object detection system using Python, PyTorch, and OpenCV using the YOLOv5 model.

Skills

- Reinforcement learning for robotic locomanipulation.
- Deployment of control policies on physical robots with embedded systems, sensors and actuators.